

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-LV-6701-07 Sheet 1 of 1	
OPL Title	Actuator Diaphragm Inspection (Fisher 667)	Discipline	Instrument		
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	Area / Unit	6700 - LP SEPARATOR SYSTEM		
Related Interlock	SEQ-6701 (SIL 1)	P&ID	Ref	TJC-LLD-PID-6701	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E	

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **actuator diaphragm inspection (fisher 667)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. A ruptured actuator diaphragm causes loss of control and a continuous air leak; periodic inspection prevents surprise failures. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Inspect the Fisher 667 diaphragm for cracks or perishing	Loop isolated / under permit
2	Check the bench set is 0.2-1.0 barg	Reference standard traceable
3	Replace the NBR/fabric diaphragm if hardened	As-found & as-left recorded
4	Re-stroke and recalibrate after any diaphragm work	Trip / control action verified
5	Confirm the spring range still gives full 38 mm travel	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- A ruptured actuator diaphragm causes loss of control and a continuous air leak; periodic inspection prevents surprise failures.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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